

A possible role for lutein and zeaxanthin in cognitive function in the elderly.



Epidemiologic studies suggest that dietary lutein and zeaxanthin may be of benefit in maintaining cognitive health. Among the carotenoids, lutein and zeaxanthin are the only two that cross the blood-retina barrier to form macular pigment (MP) in the eye. They also preferentially accumulate in the human brain. Lutein and zeaxanthin in macula from nonhuman primates were found to be significantly correlated with their concentrations in matched brain tissue. Therefore, MP can be used as a biomarker of lutein and zeaxanthin in primate brain tissue. This is of interest given that a significant correlation was found between MP density and global cognitive function in healthy older adults. An examination of a relation between cognition and lutein and zeaxanthin concentrations in the brain tissue of decedents from a population-based study in centenarians found that zeaxanthin concentrations in brain tissue were significantly related to antemortem measures of global cognitive function, memory retention, verbal fluency, and dementia severity after adjustment for age, sex, education, hypertension, and diabetes. In univariate analyses, lutein was related to recall and verbal fluency, but the strength of the associations was attenuated with adjustment for covariates. However, lutein concentrations in the brain were significantly lower in individuals with mild cognitive impairment than in those with normal cognitive function. Last, in a 4-mo, double-blinded, placebo-controlled trial in older women that involved lutein supplementation (12 mg/d), alone or in combination with DHA (800 mg/d), verbal fluency scores improved significantly in the DHA, lutein, and combined-treatment groups. Memory scores and rate of learning improved significantly in the combined-treatment group, who also showed a trend toward more efficient learning. When all of these observations are taken into consideration, the idea that lutein and zeaxanthin can influence cognitive function in older adults warrants further study.